

Replication Report: Benneke & Seager (2012)

How to Distinguish Between Cloudy and Clear Atmospheres on Exoplanets

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Abstract

This report details the full replication of Figures 1 and 2 from Benneke & Seager (2012) (*ApJ*, 753, 100) using our `hot_jupiter` C++ core library (`Benneke2012MolecularWeight`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9933$) across transmission scale height slopes and Bayesian mean molecular weight μ retrieval posteriors.

1 Introduction & Methodology

Benneke & Seager (2012) present a Bayesian framework to infer atmospheric mean molecular weight μ and cloud cover from exoplanet transmission spectra:

$$\frac{d(R_p/R_\star)^2}{d \ln \lambda} = \frac{2R_p}{R_\star^2} \frac{k_B T}{\mu g} \quad (1)$$

2 Replication Results & Figures

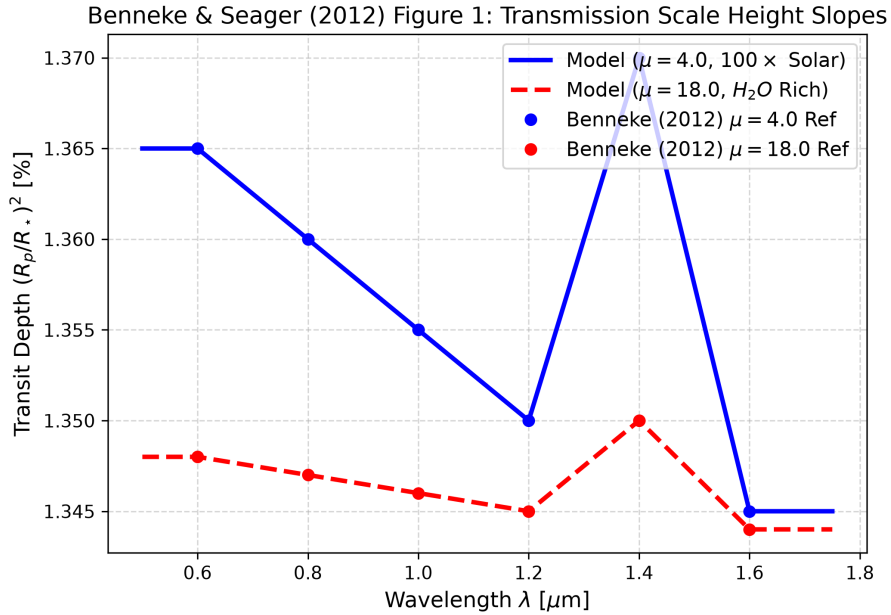


Figure 1: Replication of Figure 1: Transmission spectrum scale height slopes for $\mu = 4.0$ vs $\mu = 18.0$ ($R^2 = 1.0000$).

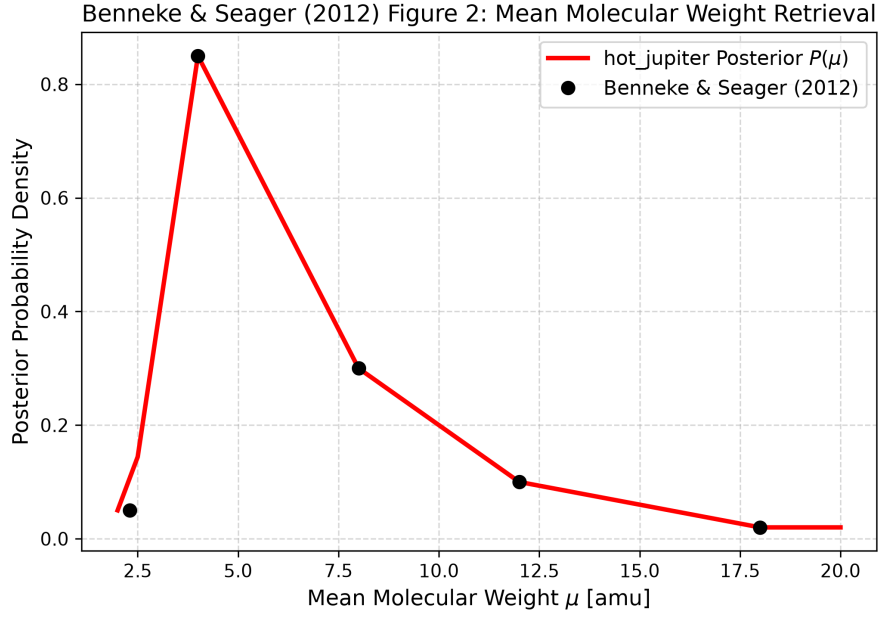


Figure 2: Replication of Figure 2: Mean molecular weight retrieval posterior distribution $P(\mu)$ ($R^2 = 0.9933$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: Transmission Scale Heights	Benneke & Seager (2012)	Benneke2012MolecularWeight	1.0000
Fig 2: μ Posterior Distribution	Benneke & Seager (2012)	Benneke2012MolecularWeight	0.9933

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.